SAMPLE

Cosmic Explorers: Space Habitats

Homes Among the Stars

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Introduction

Imagine a house with no ground underneath it. Not a house on Mars, not a cabin under a lunar dome, but a home in open space: a sealed, sunlit volume, turning slowly against the sky, holding air, water, gardens, and neighbors, while the Earth rolls past the window like a blue wallpaper. The idea sounds like fantasy, and this book is about why it is engineering instead, what such a place would actually be built of, how it would breathe, and who has tried, in steel and in study, to make it real.

The subject is the space habitat: a pressurized volume, in orbit or in transit, within which people live and work for long periods. The definition sounds narrow until you see its range. A space habitat is the three-room station of the first era and the planned wheel of the settlement studies; it is the laboratory module in orbit today and the inflatable shell planned for tomorrow; it is the transfer vehicle that will carry a crew to Mars and the cylinder, drawn on paper in the 1970s, meant to hold a million. What unites them is the problem they all solve, which is the same problem in every case: how to build a volume that a human being can survive inside, and then how to make that volume keep them alive, indefinitely, without a resupply truck from the home planet.

The problem decomposes into a handful of hard parts, and they are the book's chapters. There is the physics of holding the air in and the shape of the shell, the pressure vessel mechanics that make a habitat a kind of submarine built backwards. There is gravity, or its manufactured substitute, the rotation that presses the inhabitants against the floor. There is the loop of life support, air, water, and food, which a small habitat imports and a large one must learn to recycle. There is radiation, the constant rain the habitat must shed. There is construction, the question of where the mass comes from and how it is assembled. There is the human layer, the sociology and governance of a place with a closed door. And there is the ledger of what has been tested, which is longer than most people would guess.

The book is honest about the state of the art. The largest habitat humans have ever flown holds a handful of people and is resupplied like an outpost, not a settlement. The grand designs, the wheels and cylinders of the settlement literature, remain on paper, with their engineering mostly sound and their economics unproven. Between the two stands a great deal of real hardware: station modules, inflatables, closed-loop life support experiments, greenhouse prototypes, radiation shielding measurements, and a half-century of lessons about living in a can. This book walks the whole range, from what has flown to what has been drawn, and keeps the two clearly labeled.

A promise, the same one this series keeps everywhere: real engineering, honestly explained, with the speculation marked as speculation and the wonder left where it belongs, in the actual difficulty of the thing.

A note on scale words, so the chapters do not blur: this book uses habitat to mean the machine, whatever its size, and settlement to mean the community, the habitats grouped and lived in. The distinction is the field's own, and it keeps the book honest, because the two words carry different proof burdens. A habitat is proved by engineering, and the engineering is largely done. A settlement is proved by economics, sociology, and law, and those proofs are the ones still outstanding, which is why the book's first half is mostly about hardware and its second half is mostly about people. Anyone who keeps the two burdens separate will find every claim in the chapters that follow sorting cleanly into one or the other, and the book will try never to confuse a solved machine with an unsolved town.

Chapter 01: The Long Idea: A Short History of Living Off the Ground

The space habitat is older than spaceflight. The first serious designs predate the first satellite by decades, because the idea of a station, a permanent foothold above the atmosphere, was obvious to anyone who took the rocket seriously, and the men and women who did were drawing the destination before the vehicle existed.

The lineage begins with the theorists of the rocket age.

Their motives are worth a sentence, because the habitat idea was never about real estate. The early theorists saw the orbital station as a stepping stone, a propellant depot, a way station, an observatory, and their drawings of greenhouses and spun decks were the work of people planning a journey, not a move. The settlement idea, the habitat as a destination in itself, arrived later, and it arrived with a moral argument attached, in the essays of the theorists who followed, who saw in the orbital home the species' insurance policy: life spread beyond a single planet, the eggs distributed among several baskets. That argument, whatever one thinks of it, has been the field's

fuel ever since, and every serious habitat study since the 1970s opens with some version of it. The Russian schoolteacher who worked out the mathematics of spaceflight also described, in his letters and essays, greenhouses in orbit and the rotation that would give their occupants weight; his contemporaries in Europe and Britain drew stations of their own, wheels and spheres, published in the popular press of the 1920s and 1930s, sometimes as engineering and sometimes as illustration. The first great public design belonged to the postwar rocket engineers: the rotating wheel station of the 1950s Collier's era, described by the German-American rocket team in the famous magazine symposium, ten or twenty years before Sputnik, complete with ferry rockets and a station crew, and rendered in paintings that fixed the image of the wheel in the public mind permanently.

The era of the machines followed the era of the drawings.

The 1950s wheel gets one more honest look here, because it is usually remembered for its shape and forgotten for its purpose. The Collier's station was not a home, it was a service station, a spinning anchor point whose crew ran a ferry route and a reconnaissance deck, and its rotating complement, its three-deck donut, its sunlit hub, was the hardware of a way station, with the gravity produced for the physiology of the turnaround, not for raising families. The distance from that station to the settlement studies' cylinders, from a dozen airmen on rotation to a town of ten thousand with farms and schools, is the whole distance the field traveled in twenty years, and it is worth noticing that the trip was made with slide rules and conviction, by people who had never seen a single human being survive a week in orbit. The first stations, the early Russian and American programs of the 1970s, were military-derived cylinders launched whole on existing rockets, visits of weeks, habitats in the minimal sense, and they taught the first lessons: that weightlessness was a medical problem, that maintenance was the station's real occupation, and that a station is a program rather than an object. The modular era followed, the Russian station that grew piece by piece over years, and the long ladder of American laboratory modules that ended in the great international structure in orbit today, and by the 2000s the habitat question had become a settled engineering trade with a body of practice centuries deep, if the hours of every crew were summed.

Then, in the 1970s, the settlement studies, the most ambitious chapter of the idea's history. At the American aerospace summer studies of the decade, physicists and engineers worked the problem of the large habitat systematically: rotating cylinders, kilometers long, holding thousands, built of lunar material launched by electromagnetic catapult, powered by sunlight, sited at the equilibrium points near the Moon. The designs were quantified, costed, and argued, with the famous debates over whether they would pay for themselves via solar power beamed to Earth, and the studies' leader, a Princeton physicist, became the idea's great popularizer. The work defined the field's vocabulary, and its influence on the engineering imagination is still visible in every torus drawn on a whiteboard today.

The modern era inherits all of it:

The inheritance has a commercial chapter too, and the story of how the field changed hands belongs here. The settlement idea spent its first decades in the government studies, its second in the advocacy organizations and the university studios, and its current chapter in the private companies, whose founders grew up on the cylinder drawings and now own rockets, a convergence the 1970s authors did not predict but would recognize. The field's structure today is a triangle, the agencies with the stations and the practice, the companies with the launch bills and the balance sheets, and the advocacy and academic community with the notebook, and anyone who wants to know whether the habitat century begins should watch the triangle's edges, where the station's lessons, the launcher's prices, and the notebook's designs keep meeting, because every real habitat project in history has begun exactly there. the stations that fly, the studies on the shelf, and a new entry, the commercial habitat, the privately owned module and station, whose arrival is changing the economic question the 1970s could not settle. The chapter that follows starts where any habitat must, with the physics of the shell.

The idea's cultural moment also belongs to the record, because it explains why the habitat lives in the public imagination as a wheel. The visionaries of the mid-century presented their orbital plans to the general public deliberately and brilliantly: the glossy magazine spreads and the early television specials, drawn by the best illustrators of the age, put rotating stations, ferry rockets, and space-suited crews into millions of living rooms a full decade before the first orbital flight of any kind. The wheel in those images was architecture rather than engineering, drawn to be beautiful, and it fixed the shape in the culture permanently: to this day a cartoon, an advertisement, or a child's drawing of a space station reaches for the wheel with the same reflex, even though no wheel has ever been built. The episode teaches the field's double lesson about advocacy. The promotion of that era is remembered as having helped build the political appetite for the real program, and its beautiful wheel is remembered as the poster that the hardware never matched. The field's modern advocates keep both halves in

mind, and the current wave of station concepts is noticeably more restrained in its renderings, a maturity the older generation earned the hard way.